You have correctly answered 102 of 125 questions

Question 1 of 125

What is the range of values for an 8-bit signed binary number in 2's complement format?

- (selected) -128 to 127
- 0 to 255
- -127 to 128
- -255 to 255

An 8-bit signed binary number in 2's complement format ranges from -128 to 127.

Correct

You have correctly answered 102 of 125 questions

Question 2 of 125

Clock skew can affect both setup and hold times in a synchronous design.

- (selected) True
- False

Positive clock skew affects hold timing, while negative clock skew affects setup timing.

Correct

You have correctly answered 102 of 125 questions

Question 3 of 125

The manufacturing process for a wafer typically takes 10-15 weeks.

- (selected) True
- False

It typically takes 10-15 weeks to process a wafer in a leading-edge fab.

Correct

You have correctly answered 102 of 125 questions

Question 4 of 125

Select all the components that are part of an FSM implementation: (Select all that apply)

- [x] Output logic
- [x] Next state logic
- [ ] Memory array
- [x] State register

An FSM implementation includes next state logic, output logic, and a state register.

Correct

You have correctly answered 102 of 125 questions

Question 5 of 125

What is the primary purpose of the specification in the IC design flow?

- (selected) To provide an explicit set of requirements for the design
- To create the RTL
- To define the micro-architecture
- To perform physical verification

The specification is an explicit set of requirements that guides the design process.

Correct

You have correctly answered 102 of 125 questions

Question 6 of 125

What is the main advantage of a Carry Select Adder (CSA) over a Ripple Carry Adder?

- It uses fewer gates
- (selected) It is faster
- It consumes less power
- It is simpler to design

A Carry Select Adder is faster because it pre-computes sums for both possible carry-in values and uses a multiplexer to select the correct result.

Correct

What are the three phases of the IC design flow?

- ( ) Logical, Physical, Electrical
- ( ) Design, Verification, Implementation
- (•) System, Logical, Physical
- ( ) Specification, Micro-architecture, RTL

The IC design flow is divided into three phases: System, Logical, and Physical.

Correct

The placement process has no effect on the timing and area of the design.

- ( ) True
- (•) False

Placement significantly affects the timing and area of the design by determining the specific locations of discrete components.

Correct

You have correctly answered 102 of 125 questions

Question 9 of 125

What does a decoder typically do?

- ( ) Converts a one-hot input into a binary output.
- ( ) Selects one of several inputs based on a control signal.
- (•) Converts a binary input into a one-hot output.
- ( ) Inverts the input signal.

A decoder typically converts a binary input into a one-hot output, where only one output line is active at a time.

Correct

You have correctly answered 102 of 125 questions

Question 10 of 125

Which of the following is a legal Verilog identifier?

- ( ) 16bitbus
- ( ) unit-32
- (•) abc$
- ( ) $abc

Identifiers can include letters, digits, dollar signs, and underscores, but must start with a letter or underscore.

Correct

You have correctly answered 102 of 125 questions

Question 11 of 125

Select all valid types of breakpoints used in interactive debugging: (Select all that apply)

- [x] Time-based breakpoints
- [ ] Power-based breakpoints
- [ ] Frequency-based breakpoints
- [x] Value-based breakpoints

Time-based breakpoints stop the simulation at a specific time, and value-based breakpoints stop the simulation when specific signals take on desired values.

Correct

You have correctly answered 102 of 125 questions

Question 12 of 125

What is the purpose of setting breakpoints in interactive debugging?

- (x) To stop the simulation at specific points
- ( ) To generate test vectors
- ( ) To optimize the code
- ( ) To compile the code

Breakpoints are used to pause the simulation at specific points to examine the state of the design.

Correct

You have correctly answered 102 of 125 questions

Question 13 of 125

What is the basic building block of a Verilog design?

- Parameter
- Wire
- Register
- Module (selected)

The module is the fundamental building block in Verilog, used to describe the design hierarchy.

Correct

You have correctly answered 102 of 125 questions

Question 14 of 125

The MSB in a signed binary number determines the sign of the number.

- True (selected)
- False

The MSB in a signed binary number indicates the sign, with 0 for positive and 1 for negative.

Correct

You have correctly answered 102 of 125 questions

Question 15 of 125

Which of the following are steps involved in the IC manufacturing process? (Select all that apply)

- [x] Taping out
- [ ] Software synthesis
- [x] Physical verification
- [x] Functional verification

Taping out, functional verification, and physical verification are key steps in the IC manufacturing process.

Correct

You have correctly answered 102 of 125 questions

Question 16 of 125

Modifying the RTL is usually the first step taken to fix timing violations.

- ( ) True
- (•) False

Modifying the RTL is usually the last resort because it has a significant impact on the schedule.

Correct

You have correctly answered 102 of 125 questions

Question 17 of 125

How is a negative number represented in 2's complement format?

- ( ) By inverting the bits and subtracting 1
- (x) By inverting the bits and adding 1
- ( ) By inverting the bits
- ( ) By adding 1 to the positive number

In 2's complement format, a negative number is obtained by inverting the bits of the positive number and adding 1.

Correct

You have correctly answered 102 of 125 questions

Question 18 of 125

The quality of the RTL directly affects the end product in terms of time, effort, and money.

- (x) True
- ( ) False

The quality of the RTL directly affects the end product in terms of time, effort, and money.

Correct

You have correctly answered 102 of 125 questions

Question 19 of 125

The gate of a MOSFET is made of polysilicon.

- (•) True
- ( ) False

The gate of a MOSFET is typically made of polysilicon.

Correct

You have correctly answered 102 of 125 questions

Question 20 of 125

In a Moore FSM, the outputs can change immediately with the inputs.

- ( ) True
- (•) False

In a Moore FSM, the outputs depend only on the current state and do not change immediately with the inputs.

Correct

You have correctly answered 102 of 125 questions

Question 21 of 125

Which of the following is true about the propagation delay (tpd) in combinational logic?

- ( ) It is the delay caused by the setup and hold times of a flip-flop.
- ( ) It is the time taken for the clock signal to propagate through the circuit.
- ( ) It is the delay caused by clock skew.
- (•) It is the time taken for the output to stabilize after the input changes.

Propagation delay is the time interval between the change in input and the corresponding change in output in combinational logic.

Correct

You have correctly answered 102 of 125 questions

Question 22 of 125

What is the typical size of a silicon chip (IC)?

- (•) $100mm^2$
- ( ) 13.4mm x 13.4mm
- ( ) 50mm x 50mm
- ( ) $300mm^2$

A typical silicon chip is approximately $180mm^2$, which translates to about 13.4mm x 13.4mm.

Incorrect

By convention, what is the Most-Significant Bit (MSB) in a binary bus?

- ( ) The middle bit
- ( ) The least significant bit
- ( ) The right-most bit
- (•) The left-most bit

The MSB is the left-most bit in a binary bus, representing the highest value.

Correct

Which process involves inserting buffers in the clock tree of a digital design?

- ( ) Routing
- (•) Clock tree synthesis (CTS)
- ( ) Placement
- ( ) Floor-planning

CTS involves inserting buffers in the clock tree to ensure proper clock distribution

Correct

What is the main goal of floor-planning in the physical design process?

- ○ To place standard cells in specific locations
- ○ To insert buffers in the clock tree
- ○ To connect the pins of standard cells
- ◉ To lay out the physical partitions of a design

Floorplanning involves laying out the physical partitions of a design to determine the size and connectivity of each partition.

Correct ×

Which of the following is a key challenge throughout the implementation phase of digital IC design?

- ○ Writing the specification
- ○ Creating the micro-architecture
- ◉ Managing power consumption
- ○ Layout placement

Managing power consumption is a key challenge in the implementation phase, requiring sophisticated strategies.

Correct ×

You have correctly answered 102 of 125 questions

Question 27 of 125

The propagation delay of a logic gate is the time it takes for the output to change after the input changes.

- (•) True
- ( ) False

Propagation delay is the time interval between the change in input and the corresponding change in output.

Correct

You have correctly answered 102 of 125 questions

Question 28 of 125

A FIFO memory reads data in the order it was written.

- (•) True
- ( ) False

FIFO (First-In-First-Out) memory reads data in the same order it was written.

Correct

What is the purpose of the silicon dioxide (SiO2) layer in a MOSFET?

- ○ To conduct electricity
- ◉ To act as an insulating layer
- ○ To provide structural support
- ○ To increase the transistor's speed

The SiO2 layer in a MOSFET acts as an insulator between the gate and the body of the transistor.

Correct ×

Which of the following statements about P-type semiconductors are true? (Select all that apply.)

- [x] Have more valence electrons
- [ ] Have more valence holes
- [x] Doped with Group III elements
- [x] Conduct electricity using positive charge carriers

P-type semiconductors are doped with Group III elements, have more valence holes, and conduct electricity using positive charge carriers.

Incorrect ×

You have correctly answered 102 of 125 questions

Question 31 of 125

Which phase involves converting the high-level design specification into detailed design components and blocks?

- ○ Physical phase
- ○ Verification phase
- ◉ Logical phase
- ○ System phase

The logical phase involves developing the RTL and detailed design components and blocks.

Correct

You have correctly answered 102 of 125 questions

Question 32 of 125

The always procedural block is non-synthesizable and used only for testbenches.

- ○ True
- ◉ False

The always procedural block is synthesizable and used to describe hardware that reacts to inputs. The initial procedural block is non-synthesizable and used only for testbenches.

Correct

You have correctly answered 102 of 125 questions

Question 33 of 125

Which verification process checks the logical equivalence of the design using mathematical proofs?

- ◯ Logic simulation
- ◯ System simulation
- ◯ Physical verification
- ◉ Formal verification

Formal verification checks the logical equivalence of the design.

Correct

You have correctly answered 102 of 125 questions

Question 34 of 125

System simulation is performed at the RTL level.

- ◯ True
- ◉ False

System simulation is performed at the behavioral or system level, not at the RTL level.

Correct

Select all the steps involved in the back-end or physical design process:
(Select all that apply)

- [ ] Static timing analysis
- [x] Floor-planning
- [x] Logic synthesis
- [x] Placement

The back-end or physical design process includes steps such as floorplanning, placement, and static timing analysis.

Incorrect

What is the prime advantage of a Ripple-Carry adder implementation?

- ( ) Higher power consumption
- (•) Smaller area
- ( ) Faster speed
- ( ) More complex design

N-bit addition is implemented by cascading N full adders in a ripple carry adder architecture.

Correct

You have correctly answered 102 of 125 questions

Question 37 of 125

Which of the following are levels of abstraction supported by HDLs? (Select all that apply.)

- [x] Switch level
- [x] Register Transfer Level (RTL)
- [x] Behavioral level
- [x] Gate level

HDL supports multiple levels of abstraction, including behavioral, RTL, gate, and switch levels.

Correct

You have correctly answered 102 of 125 questions

Question 38 of 125

What is the primary goal of functional verification?

- (•) To demonstrate the functional correctness of a design
- ( ) To reduce the power consumption of the design
- ( ) To optimize the design for performance
- ( ) To minimize the area of the design

The primary goal of functional verification is to ensure that the design functions correctly according to the specifications.

Correct

You have correctly answered 102 of 125 questions

Question 39 of 125

In Gray code, multiple bits change state from one value to the next.

- True
- False

In Gray code, only one bit changes state from one value to the next, ensuring safer transitions.

Correct

You have correctly answered 102 of 125 questions

Question 40 of 125

Logical shifts are used for signed arithmetic operations.

- True
- False

Logical shifts are used for unsigned arithmetic operations, while arithmetic shifts are used for signed arithmetic to maintain the sign bit.

Correct

You have correctly answered 102 of 125 questions

Question 41 of 125

Which of the following statements about Binary Coded Decimal (BCD) are true? (Select all which apply.)

- [ ] Arithmetic on multi-digit values is simple
- [x] It is used for clock and counter displays
- [ ] Values 1010 to 1111 are used
- [x] Each decimal digit is stored as 4 bits

BCD represents each decimal digit as 4 bits and is used for applications like clock and counter displays. Values 1010 to 1111 are not used.

Correct ×

You have correctly answered 102 of 125 questions

Question 42 of 125

What is the function of the -access rwc option in the xrun command?

- ○ To enable line debugging
- ◉ To enable read, write, and connectivity access to design objects
- ○ To set the simulation time scale
- ○ To specify the input files

The -access rwc option allows read, write, and connectivity access to the design objects, which is necessary for probing signals and displaying waveforms.

Correct ×

You have correctly answered 102 of 125 questions

Question 43 of 125

Which of the following is NOT a basic logic gate?

- AND
- OR
- MUX
- NOT

MUX (Multiplexer) is a combinational building block, not a basic logic gate. Basic gates include AND, OR, and NOT.

Correct

You have correctly answered 102 of 125 questions

Question 44 of 125

The process node is a fixed physical measure of transistor gate length.

- True
- False

The process node used to be a physical measure of transistor gate length, but now it is more of a marketing term.

Correct

You have correctly answered 102 of 125 questions

Question 45 of 125

Which element is commonly used to dope N-type semiconductors?

- ○ Boron
- ◉ Arsenic
- ○ Silicon
- ○ Germanium

Arsenic, a Group V element, is commonly used to dope N-type semiconductors, providing extra electrons.

Correct

You have correctly answered 102 of 125 questions

Question 46 of 125

In a forward-biased PN junction, current flows easily.

- ◉ True
- ○ False

In a forward-biased PN junction, the current flows easily from the P-type to the N-type material.

Correct

What happens when a "0" voltage is applied to the gate of an nMOS transistor?

- (x) No current flows
- ( ) The transistor heats up
- ( ) Current flows from source to drain
- ( ) Current flows from drain to source

When a "0" voltage is applied to the gate of an nMOS transistor, holes are attracted to the gate and stopped by the SiO2 layer, preventing current flow.

Correct

Which of the following are key factors in optimizing IC design for profitability? (Select all which apply.)

- [x] Power
- [x] Area
- [ ] Security
- [x] Performance

Power, performance, area, and security are all critical factors in optimizing IC design for profitability.

Incorrect

You have correctly answered 102 of 125 questions

Question 49 of 125

The effective clock period can be reduced by clock skew.

- (selected) True
- False

Clock skew can reduce the effective clock period, especially if there is negative skew, where the clock arrives at the target register before the source register.

Correct

You have correctly answered 102 of 125 questions

Question 50 of 125

Which of the following is true about synchronous memory?

- (selected) It introduces pipeline delays
- It does not use clocks
- It is faster than asynchronous memory
- It cannot be used with processors

Synchronous memory introduces pipeline delays due to internal register banks.

Correct

You have correctly answered 102 of 125 questions

Question 51 of 125

High-level decisions in the design flow are made at the specification level.

- (x) True
- ( ) False

High-level decisions are made at the specification level, affecting the system and its environment.

Correct

You have correctly answered 102 of 125 questions

Question 52 of 125

What is Moore's Law?

- ( ) The transistor density of an IC doubles every two years
- ( ) The power consumption of ICs halves every two years
- ( ) The size of ICs halves every two years
- (x) The cost of IC manufacturing doubles every two years

Moore's Law states that the transistor density of an IC doubles approximately every two years.

Incorrect

The RTL phase is the final step before physical verification.

- ○ True
- ◉ False

The RTL phase is followed by logic synthesis and other steps before physical verification.

Correct

Division can be implemented using a shift-subtract array.

- ◉ True
- ○ False

Division can be implemented using a shift-subtract array, which repeatedly subtracts the divisor from the dividend and shifts the result.

Correct

You have correctly answered 102 of 125 questions

Question 55 of 125

The reg data type can be declared as signed in Verilog-2001.

- ○ True
- ◉ False

Verilog-2001 allows the reg data type to be declared as signed.

Incorrect

You have correctly answered 102 of 125 questions

Question 56 of 125

One-hot encoding requires fewer state registers than binary encoding.

- ○ True
- ◉ False

One-hot encoding requires more state registers than binary encoding, as each state is represented by a unique bit.

Correct

You have correctly answered 102 of 125 questions

Question 57 of 125

Select all the methods that can be used for signed multiplication: (Select all that apply)

- [x] Using Booth's algorithm
- [ ] Using a ripple carry adder
- [x] Using a shift-adder array with sign extension
- [x] Using unsigned multiplication and adjusting the sign

Signed multiplication can be implemented using unsigned multiplication with sign adjustment, a shift-adder array with sign extension, or Booth's algorithm.

Correct

You have correctly answered 102 of 125 questions

Question 58 of 125

Which of the following is a valid way to declare a module parameter in Verilog-2001?

- ( ) defparam WIDTH = 2;
- ( ) param WIDTH = 2;
- (•) #(parameter integer WIDTH = 2)
- ( ) localparam WIDTH = 2;

Verilog-2001 allows parameters to be declared using the #(parameter integer WIDTH = 2) syntax.

Correct

You have correctly answered 102 of 125 questions

Question 59 of 125

Which of the following is a characteristic of a Mealy FSM?

- Outputs depend only on the current state
- State transitions are asynchronous
- Requires more states than a Moore FSM
- Outputs depend on the current state and inputs

In a Mealy FSM, the outputs depend on both the current state and the inputs.

Correct

You have correctly answered 102 of 125 questions

Question 60 of 125

What is the primary function of a transistor?

- To insulate circuits
- To generate electricity
- To store data
- To amplify electric signals

Transistors are used to amplify electric signals and act as digital switches.

Correct

You have correctly answered 102 of 125 questions

Question 61 of 125

Which of the following are steps involved in post-processing debug mode? (Select all that apply.)

- [x] Waveform signal comparison
- [x] Probing assertions
- [ ] Code synthesis
- [x] Driver tracing

Post-processing debug mode involves waveform signal comparison, driver tracing, and probing assertions to analyze the simulation results.

Correct

You have correctly answered 102 of 125 questions

Question 62 of 125

What is a semiconductor?

- (●) A material that conducts electricity better than insulators but not as well as conductors
- ( ) A material that does not conduct electricity at all
- ( ) A material that conducts electricity as well as conductors
- ( ) A material that only conducts electricity at high temperatures

Semiconductors have electrical conductivity between that of insulators and conductors.

Correct

You have correctly answered 102 of 125 questions

Question 63 of 125

The lithographic process uses ultraviolet light to etch patterns onto the wafer.

- ( ) True
- (•) False

The lithographic process uses ultraviolet light to etch patterns onto the photosensitive polymer on the wafer.

Incorrect

You have correctly answered 102 of 125 questions

Question 64 of 125

Select all valid ways to connect ports in a Verilog module instantiation:
(Select all that apply)

- [x] Ordered port connection
- [ ] Direct port connection
- [x] Named port connection
- [ ] Implicit port connection

Ports can be connected using named port connection or ordered port connection.

Correct

You have correctly answered 102 of 125 questions

Question 65 of 125

The ripple carry adder is faster than the carry select adder.

- ○ True
- ◉ False

The ripple carry adder is slower because the carry must propagate through each stage, whereas the carry select adder pre-computes sums for both possible carry-in values.

Correct

You have correctly answered 102 of 125 questions

Question 66 of 125

What is the purpose of static timing analysis (STA) in the design flow?

- ○ To perform physical verification
- ○ To verify the logical equivalence of the design
- ◉ To compute the expected timing of a digital circuit
- ○ To create the RTL

STA is used to compute the expected timing of a digital circuit without using SPICE simulation.

Correct

You have correctly answered 102 of 125 questions

Question 67 of 125

A multiplexer can be used to implement basic logical gates by appropriately connecting its input and control signals.

- (selected) True
- False

A multiplexer can be configured to implement various logical gates by appropriately connecting its inputs and control signals.

Correct

You have correctly answered 102 of 125 questions

Question 68 of 125

What does the term "process node" represent in IC manufacturing?

- (selected) The minimum size of physical features on an IC
- The cost of manufacturing an IC
- The speed of the IC
- The number of transistors per chip

The process node represents the minimum size of physical features that can be manufactured on an IC, though it has become more of a marketing term over time.

Correct

Booth's algorithm can be used for both signed and unsigned multiplication.

- (selected) True
- False

Booth's algorithm can be used for both signed and unsigned multiplication by encoding the multiplier to reduce the number of partial products.

Correct

What is the primary function of a FIFO memory?

- To delay data for a fixed number of clock cycles
- To provide dual-port access
- To store data in a stack
- (selected) To store data in the order it was written and read it in the same order

FIFO (First-In-First-Out) memory stores data in the order it was written and reads it in the same order.

Correct

You have correctly answered 102 of 125 questions

Question 71 of 125

The first transistor was made using silicon.

- ○ True
- ◉ False

The first transistor was made using germanium.

Correct

You have correctly answered 102 of 125 questions

Question 72 of 125

What is the first step in the implementation flow of digital IC design?

- ○ Floor-planning
- ○ Static timing analysis
- ○ Placement
- ◉ Logic synthesis

Logic synthesis is the first step in the implementation flow, where RTL is translated into a gate-level netlist.

Correct

You have correctly answered 102 of 125 questions

Question 73 of 125

Which of the following is NOT a method of verification?

- Emulation
- Synthesis
- Simulation
- Formal verification

Synthesis is the process of converting a high-level design description into a gate-level representation, not a verification method.

Correct

You have correctly answered 102 of 125 questions

Question 74 of 125

Select all valid Verilog data types: (Select all that apply)

- [x] wire
- [x] float
- [ ] integer
- [x] reg

Verilog supports integer, wire, and reg data types. Float is not a valid Verilog data type.

Incorrect

You have correctly answered 102 of 125 questions

Question 75 of 125

Timing closure is the process of iterating through synthesis, physical design, and verification to converge on the timing of a digital design.

- (●) True
- ( ) False

Timing closure involves iterating through synthesis, physical design, and verification to meet the timing requirements of the design.

Correct ×

You have correctly answered 102 of 125 questions

Question 76 of 125

The micro-architecture phase involves defining the intercommunication among design components and blocks.

- (●) True
- ( ) False

The microarchitecture phase involves defining the intercommunication among design components and blocks.

Correct ×

You have correctly answered 102 of 125 questions

Question 77 of 125

Which of the following are characteristics of semiconductors? (Select all that apply.)

- [x] Conduct better than insulators
- [ ] Conduct as well as conductors
- [x] Can be doped to change conductivity
- [ ] Do not conduct electricity at all

Semiconductors conduct better than insulators and can be doped to modify their conductivity.

Correct

You have correctly answered 102 of 125 questions

Question 78 of 125

What is the typical cost to manufacture a high-end processor wafer?

- ( ) $10,000
- ( ) $6,000
- ( ) $120,000
- (•) $50,000

The pure manufacturing cost of a high-end processor wafer is approximately $6,000.

Incorrect

What is the primary function of a memory compiler?

- (selected) To provide simulation, synthesis, and layout models for memory
- To design memory architectures
- To generate memory blocks from HDL code
- To optimize memory access timing

A memory compiler generates vendor-specific models for simulation, synthesis, and layout.

Correct

In 2's complement format, the range of an 8-bit signed binary number is 0 to 255.

- True
- (selected) False

In 2's complement format, the range of an 8-bit signed binary number is -128 to 127.

Correct

You have correctly answered 102 of 125 questions

Question 81 of 125

Flip-flops are more area-efficient than latches.

- True
- False (selected)

Latches are more area-efficient than flip-flops, but flip-flops are easier to design and implement.

Correct

You have correctly answered 102 of 125 questions

Question 82 of 125

A dual-port memory can have one port for reading and another for writing.

- True (selected)
- False

Dual-port memory can be configured with one port for reading and another for writing.

Correct

You have correctly answered 102 of 125 questions

Question 83 of 125

What is the primary reason for using binary numbers in digital circuits?

- ○ They require less power
- ◉ Circuits only discriminate between two states: 1 and 0
- ○ They are easier to read
- ○ They are faster to process

Digital circuits operate using binary states, where transistors act as switches that are either on (1) or off (0).

Correct ×

You have correctly answered 102 of 125 questions

Question 84 of 125

In a Moore FSM, the outputs are a function of:

- ○ Current state and inputs
- ○ Current state only
- ○ Previous state and inputs
- ◉ Inputs only

In a Moore FSM, the outputs depend only on the current state.

Incorrect ×

You have correctly answered 102 of 125 questions

Question 85 of 125

Moore's Law has been accurate for over 50 years.

- (●) True
- ( ) False

Moore's Law has been remarkably accurate for over 50 years, predicting the doubling of transistor density every two years.

Correct

You have correctly answered 102 of 125 questions

Question 86 of 125

The state space for a complex processor design is relatively small and manageable.

- (●) True
- ( ) False

The state space for a complex processor design is massive and unfeasible to simulate every possible scenario.

Incorrect

You have correctly answered 102 of 125 questions

Question 87 of 125

Emulation is a static verification method.

- (selected) True
- False

Emulation is a dynamic verification method that executes and debugs RTL and software on custom hardware much faster than a software simulator.

Incorrect

×

You have correctly answered 102 of 125 questions

Question 88 of 125

All nets in Verilog must be explicitly declared by default.

- (selected) True
- False

Verilog allows implicit net declarations unless overridden by the default_nettype none directive.

Incorrect

×

You have correctly answered 102 of 125 questions

Question 89 of 125

Which of the following statements about Verilog comments are true? (Select all that apply)

- [x] Block comments start with /*
- [x] Single-line comments start with //
- [ ] Block comments can be nested
- [ ] Single-line comments end with */

Single-line comments start with // and block comments start with /* and end with */. Block comments cannot be nested.

Correct

You have correctly answered 102 of 125 questions

Question 90 of 125

Which of the following memory structures is used to directly implement a stack?

- ( ) Dual-port memory
- ( ) FIFO
- ( ) Synchronous memory
- (•) LIFO

A stack is implemented using Last-In-First-Out (LIFO) memory.

Correct

You have correctly answered 102 of 125 questions

Question 91 of 125

What is the primary function of a multiplexer (MUX)?

- ○ To decode binary inputs into multiple outputs
- ○ To encode multiple inputs into a single output
- ○ To invert the input signal
- ◉ To select between different data inputs based on a control input

A multiplexer selects one of several input signals and forwards the selected input to a single output line based on the value of the control input.

Correct

You have correctly answered 102 of 125 questions

Question 92 of 125

The xrun command with the -linedebug option enables support for setting line breakpoints and single-stepping through the code.

- ◉ True
- ○ False

The -linedebug option in the xrun command enables line debugging features.

Correct

You have correctly answered 102 of 125 questions

Question 93 of 125

Select all the gates that can be formed by adding an inverter to another gate:

(Select all that apply)

- [x] XNOR
- [x] NOR
- [x] NAND
- [ ] XOR

NAND is formed by adding an inverter to an AND gate, NOR is formed by adding an inverter to an OR gate, and XNOR is formed by adding an inverter to an XOR gate.

Correct

You have correctly answered 102 of 125 questions

Question 94 of 125

Which gate is functionally complete and can be used to build all other logic gates?

- (•) NAND
- ( ) AND
- ( ) XOR
- ( ) OR

NAND gates are functionally complete, meaning all other logic gates can be constructed using combinations of NAND gates.

Correct

You have correctly answered 102 of 125 questions

Question 95 of 125

The initial procedural block loops back to the beginning upon completion.

- ○ True
- ◉ False

The initial procedural block executes once and terminates upon completion.

Correct ×

You have correctly answered 102 of 125 questions

Question 96 of 125

What is the primary difference between a net and a variable in Verilog?

- ○ Nets are always single-bit, variables can be multi-bit.
- ◉ Nets are driven by continuous assignments, variables are driven by procedural assignments.
- ○ Nets store values, variables drive values.
- ○ Nets are used for inputs, variables for outputs.

Nets represent physical connections and are driven by continuous assignments, while variables store values and are driven by procedural assignments.

Correct ×

You have correctly answered 102 of 125 questions

Question 97 of 125

Verilog identifiers are case sensitive.

- ( ) True
- (•) False

Verilog identifiers are case sensitive, meaning ABC, Abc, and abc are considered different identifiers.

Incorrect ×

You have correctly answered 102 of 125 questions

Question 98 of 125

Select all the components that are part of a flip-flop's timing characteristics:

(Select all that apply)

- [x] Setup time
- [x] Clock-to-Q delay
- [x] Hold time
- [ ] Propagation delay

Flip-flop timing characteristics include setup time, hold time, and clock-to-Q delay. Propagation delay is associated with combinational logic gates.

Correct ×

Select all valid reasons why power consumption is important in IC design:
(Select all that apply)

- [x] Lower manufacturing costs
- [x] Increased clock frequency
- [ ] Reduced cooling requirements
- [x] Longer battery life for portable devices

Power consumption is important for longer battery life and reduced cooling requirements, especially in portable devices.

Incorrect

The cost of developing a complex processor can exceed $1 billion US dollars.

- (•) True
- ( ) False

The development cost for a complex processor can be as high as $1 billion US dollars.

Correct

Which of the following statements about flip-flops are true? (Select all that apply.)

- [x] Flip-flops are used in Double-Data Rate (DDR) designs
- [x] Flip-flops are edge-triggered
- [x] Flip-flops store data on clock transitions
- [ ] Flip-flops are level-sensitive

Flip-flops are edge-triggered devices that store data on clock transitions. They are commonly used in DDR designs for high data rates.

Correct

A priority encoder outputs the value corresponding to the lowest priority input that is set.

- ( ) True
- (•) False

A priority encoder outputs the value corresponding to the highest priority input that is set.

Correct

You have correctly answered 102 of 125 questions

Question 103 of 125

The initial procedural block in Verilog loops back to the beginning upon completion.

- (•) True
- ( ) False

The initial procedural block executes once and terminates upon completion.

Incorrect ×

You have correctly answered 102 of 125 questions

Question 104 of 125

Which of the following is a characteristic of a Moore FSM?

- ( ) Requires fewer states than a Mealy FSM
- ( ) Outputs can change immediately with inputs
- ( ) State transitions are asynchronous
- (•) Outputs depend only on the current state

In a Moore FSM, the outputs depend only on the current state.

Correct ×

Verification is only about testing the functional correctness of an IC design.

- True
- False

Verification also involves optimizing the design for power, performance, and area, and ensuring it meets all requirements.

Correct

What is the main purpose of using Gray code for FSM state encoding?

- To increase the number of state registers
- To reduce the number of states
- To simplify output decode logic
- To minimize power consumption and noise

Gray code minimizes power consumption and noise by ensuring only one bit changes between state transitions.

Incorrect

You have correctly answered 102 of 125 questions

Question 108 of 125

Which of the following are measured to determine the quality of the RTL during logic synthesis? (Select all that apply.)

- [ ] Yield
- [x] Area
- [x] Power
- [x] Timing

The quality of the RTL is measured by evaluating timing, area, and power during logic synthesis.

Correct

You have correctly answered 102 of 125 questions

Question 109 of 125

Select all valid steps in the lithographic process for semiconductor manufacturing: (Select all that apply)

- [ ] Infrared exposure
- [ ] Piranha etch
- [x] Substrate cleaning
- [x] Photoresist coating

Substrate cleaning, adding adhesive, and photoresist coating are key steps in the lithographic process.

Incorrect

You have correctly answered 102 of 125 questions

What is the main function of RTL (Register Transfer Level) in the IC design flow?

- [ ] To perform physical verification
- [x] To describe the operation of a digital circuit in terms of signal flow between registers
- [ ] To create the specification
- [ ] To define the micro-architecture

RTL describes the operation of a digital circuit in terms of the flow of signals between registers and the operations performed.

Correct

You have correctly answered 102 of 125 questions

What is the primary advantage of using a registered output FSM implementation?

- [ ] Faster clock to output
- [ ] Simplifies next state logic
- [ ] Requires fewer flip-flops
- [x] Removes glitches on outputs

Registering the outputs of an FSM removes glitches by breaking the direct combinational path to the outputs.

Correct

You have correctly answered 102 of 125 questions

Question 112 of 125

Which of the following statements about binary multiplication are true?

(Select all that apply.)

- [x] It produces a product with twice the number of bits as the operands
- [x] It can be implemented using shift operations
- [ ] It requires a separate adder for each bit of the multiplier
- [x] It involves repeated addition

Binary multiplication involves repeated addition and can be implemented using shift operations. The product of two n-bit numbers is an n+m bit number.

Correct

You have correctly answered 102 of 125 questions

Question 113 of 125

Which of the following is a disadvantage of using a Mealy FSM?

- ( ) Requires more states
- ( ) State transitions are asynchronous
- (•) Outputs can glitch due to unsynchronized inputs
- ( ) Outputs depend only on the current state

A disadvantage of a Mealy FSM is that the outputs can glitch due to unsynchronized inputs, as there is a direct combinational path from inputs to outputs

Correct

You have correctly answered 102 of 125 questions

Question 114 of 125

Which IEEE standard is used for floating-point representation?

- (selected) IEEE 802.11
- IEEE 754
- IEEE 1394
- IEEE 488

IEEE 754 is the standard used for floating-point representation.

Incorrect

×

You have correctly answered 102 of 125 questions

Question 115 of 125

Which keyword is used to start a procedural block that executes at the start of simulation and loops back to the beginning upon completion?

- forever
- (selected) always
- begin
- initial

The always block starts executing at the start of simulation and loops back to the beginning upon completion.

Correct

×

You have correctly answered 102 of 125 questions

Question 116 of 125

Floating-point representation can handle both large and fractional number ranges efficiently.

- (•) True
- ( ) False

Floating-point representation is designed to efficiently handle both large and fractional number ranges

Correct

You have correctly answered 102 of 125 questions

Question 117 of 125

A 7-segment decoder converts a binary input into a pattern of lit segments for display.

- (•) True
- ( ) False

A 7-segment decoder converts a binary input into a pattern of lit segments to display numbers on a 7-segment LED display.

Correct

You have correctly answered 102 of 125 questions

Question 118 of 125

Functional verification is a continuous process that runs throughout the design flow.

- ○ True
- ◉ False

Functional verification is an ongoing process that ensures the design meets the specification requirements at each phase of the design flow.

Incorrect

You have correctly answered 102 of 125 questions

Question 119 of 125

Synchronous memories introduce pipeline delays due to internal register banks.

- ◉ True
- ○ False

Synchronous memories have internal register banks that introduce pipeline delays.

Correct

You have correctly answered 102 of 125 questions

Question 120 of 125

Which of the following are included in a design specification? (Select all that apply.)

- [x] Detailed description of the design
- [x] Block-level interfaces
- [ ] Verification testbenches
- [x] Performance targets

A design specification includes block-level interfaces, performance targets, and a detailed description of the design.

Correct

You have correctly answered 102 of 125 questions

Question 121 of 125

The implementation flow starts with the micro-architecture and progresses through various representations.

- (•) True
- ( ) False

The implementation flow starts with RTL and progresses through various representations.

Incorrect

You have correctly answered 102 of 125 questions

Question 122 of 125

What is the hexadecimal representation of the binary value 8'b11010011?

- 8'hC3
- 8'hB3
- 8'hD3
- 8'hA3

The binary value 11010011 converts to the hexadecimal value D3.

Incorrect ×

You have correctly answered 102 of 125 questions

Question 123 of 125

N-type semiconductors have more positive charge carriers.

- True
- False

N-type semiconductors have more negative charge carriers (electrons).

Correct ×

You have correctly answered 102 of 125 questions

Question 124 of 125

Which of the following are characteristics verified during the functional verification process? (Select all which apply.)

- [ ] Area
- [x] Functionality
- [x] Timing
- [x] Power consumption

Functional verification checks the functionality and timing of the design to ensure it meets the specification requirements.

Incorrect

You have correctly answered 102 of 125 questions

Question 125 of 125

Flip-flop designs are more area-efficient than latch-based designs.

- ( ) True
- (•) False

Latch-based designs are more area-efficient but are more difficult to design compared to flip-flop designs.

Correct